

Revisiting China's Energy Poverty: Machine Learning Insights into Key Predictors and Nonlinear Relationships

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Abstract

This study examines the determinants of multidimensional energy poverty in China from 2012 to 2020 using an integrated approach that combines econometric analysis with machine learning techniques. Using data from the China Family Panel Survey, we constructed a Multidimensional Energy Poverty Index that captures both accessibility and affordability. Traditional regression analysis identified key predictors, including income level, education, family size, and hukou status, with urban households experiencing significantly lower levels of energy poverty. To address the limitations of linear models, we applied extreme gradient boosting and interpreted the results using Shapley Additive Explanations. This approach reveals that household income, family size, education, and hukou status are the most important predictors of energy poverty. Importantly, the results show complex nonlinear relationships that the linear regression model fails to fully capture, providing a deeper understanding of how these factors contribute to energy poverty. These insights provide robust, data-driven guidance for policymakers seeking to mitigate energy poverty in China through income support, educational programs, and rural infrastructure development.

Keywords: multidimensional energy poverty, determinants of energy poverty, explainable machine learning.

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1. Introduction

As the world's largest developing country, China has made significant progress in reducing household energy poverty by expanding access to modern energy services, particularly electricity and clean cooking fuels. Achieving universal electricity coverage in 2015 established China as a global leader in electrification. However, many households, especially in rural and underdeveloped regions, continue to rely on traditional biomass (e.g., firewood or straw) or coal for essential cooking and heating needs (Hong et al., 2022; Jiang et al., 2020; Zhang et al., 2019; Zheng et al., 2014). This dependence on polluting fuels poses serious health risks (Lim et al., 2012; Zhang et al., 2019) and highlights ongoing challenges in achieving inclusive and sustainable energy transitions.

Given these persistent issues, a comprehensive measure of energy poverty in China is necessary. The conventional approach to measuring energy poverty dates back to Boardman (1991), who classified households as energy poor if their energy expenditures exceeded 10% of their income. Following this framework, various energy poverty indices have been proposed (Barnes et al., 2011; Foster et al., 2000; Mirza & Szirmai, 2010; Pachauri et al., 2004). However, relying on a single indicator fails to capture the multifaceted nature of energy poverty (Nussbaumer et al., 2012; Okushima, 2017), which encompasses energy consumption, cost, and efficiency—factors particularly relevant in developing countries (International Energy Agency, 2002) such as China. To address this limitation, researchers have developed multidimensional indices; with a prominent example being the multidimensional energy poverty index (MEPI) proposed by Nussbaumer et al. (2012). This index accounts for energy costs, efficiency, household income, and access to essential facilities enabled by energy. While several studies have applied MEPI to measure energy poverty in China (Wang et al., 2023; Zhang et al., 2019), those spanning extended periods occasionally face measurement

inaccuracies when certain indicators are not consistently surveyed each year (Li & Ma, 2023; Wang & Lin, 2022).

Effectively addressing the remaining energy poverty challenges requires a thorough understanding of its predictors. In China, household energy consumption varies significantly due to factors such as income, geographical location, and regional climatic conditions (Lin & Wang, 2020). This heterogeneity highlights the need to analyze the specific determinants of energy poverty (Du et al., 2022; Huang et al., 2022; Xie et al., 2022). A clearer understanding of these factors is essential for designing targeted policies and interventions that address the diverse causes of energy poverty in China.

Accordingly, this study investigates the following research questions: What are the key predictors of multidimensional energy poverty in China, and how do these predictors relate to energy poverty in a nonlinear context? Many studies have used conventional regression approaches to examine the determinants of energy poverty (Gu, 2023; Hong et al., 2022; Huang et al., 2022; Li & Ma, 2023; Lin & Zhao, 2021; Luan et al., 2023; Sharma et al., 2019; Shi et al., 2023; Wang et al., 2022; Wang & Lin, 2022; Yan et al., 2024; Yin et al., 2023). However, these studies are limited by their reliance on models that fail to account for nonlinear relationships between energy poverty and its predictors. To address this gap, this study applies machine learning techniques to explore these relationships. Although previous research has used machine learning to analyze energy poverty (Abbas et al., 2022; Al Kez et al., 2024; Balkissoon et al., 2024; Dalla Longa et al., 2021; Spandagos et al., 2023; van Hove et al., 2022), it has primarily focused on building predictive models rather than examining the complex, nonlinear mechanisms underlying these relationships. By employing the explainable machine learning technique of Shapley Additive Explanations (SHAP) (Lundberg & Lee, 2017), this study expands on previous work by uncovering intricate nonlinear interactions, providing a more nuanced understanding of energy poverty in China.

This study makes three key contributions. First, it constructs a comprehensive MEPI for China using data from the 2012–2020 China Family Panel Survey (CFPS), capturing the multidimensional nature of household energy poverty. Second, it identifies key predictors of MEPI using advanced machine learning methods, contributing to methodological advancements in energy poverty research. Third, and most importantly, it reveals the complex marginal effects of these predictors on MEPI within a nonlinear framework, offering policymakers deeper insights for designing targeted interventions to alleviate energy poverty.

The remainder of this paper is structured as follows. Section 2 describes the methodology for measuring MEPI and introduces the data used for analysis. Section 3 presents the conventional econometric model and machine learning methods employed to identify MEPI predictors. Section 4 discusses the findings, including regression results, feature importance scores, and the marginal effects of key predictors. Finally, Section 5 concludes the paper, summarizes the main findings and policy implications.

2. Data and Measurement

2.1. Definition of Household Energy Poverty

Household energy poverty refers to the lack of access to, or the inability to afford sufficient, secure, clean, and environmentally sustainable modern energy (Charlier et al., 2019; Dogan et al., 2022; Reddy et al., 2000). This definition incorporates two key dimensions: accessibility and affordability of modern energy. Household energy poverty emphasizes the inability to access modern energy sources. Among its dimensions, accessibility is the most critical, as it determines whether households can use modern energy infrastructure, regardless of their ability to afford it. Accessibility primarily depends on the availability of energy infrastructure (Li et al., 2025). For example, households without access to a natural gas pipeline may rely on firewood or coal for cooking. Affordability is another key aspect of household energy poverty. While the expansion of energy infrastructure increases access, some households may still

struggle to afford modern energy. This financial burden, defined as energy affordability, remains a significant challenge. For instance, despite China achieving universal electricity coverage in 2015, insufficient electricity use persists in some areas due to affordability constraints (Li et al., 2025; Zhang et al., 2019). To capture the full extent of household energy poverty, this study employs an MEPI that combines both accessibility and affordability dimensions.

2.2. Data

We utilize the CFPS, a nationally representative panel dataset collected biennially by the Institute of Social Science Survey at Peking University. The CFPS provides detailed information on household energy consumption, making it particularly useful for constructing the MEPI. The dataset spans from 2010 to 2022, and for this study, we use data from 2012, 2014, 2016, 2018, and 2020 to ensure consistency in MEPI construction.

2.3. Measurement of MEPI

This study employs a comprehensive approach to measure household energy poverty, integrating both the accessibility and affordability dimensions using CFPS data from 2012 to 2020. Specifically, the accessibility dimension was assessed using three indicators: reliance on firewood or straw as the primary cooking fuel¹, use of coal as the main cooking fuel², and use of unsanitized water sources (e.g., river or lake water, wells, rainwater, cellar water, or spring

¹ The firewood is measured by the question “What kind of fuel does your family normally use for cooking?” in the CFPS Questionnaire. If the primary cooking fuel is firewood or straw, we assign firewood=1; otherwise, firewood=0. For households primarily using firewood or straw, their fuel sources are typically free. Firewood is collected from forests, while straw is available as a byproduct after harvesting wheat. This indicator captures the energy poverty issues faced by rural, agriculture-based households in China. In these rural areas, energy infrastructure is underdeveloped, leaving rural households unable to access modern energy, such as natural gas and liquid gas. This reflects the challenge of household energy poverty from the perspective of accessibility.

² The coal is captured by the question “What kind of fuel does your family normally use for cooking?” in the CFPS Questionnaire. If the primary cooking fuel is coal, we assign coal=1; otherwise, coal=0. For households that primarily use coal for cooking, this fuel source is typically low-cost. For instance, briquette coal is usually priced at around 0.5 RMB per unit, with one briquette lasting approximately three days. This cost is significantly lower than that of electricity or natural gas.

water) for cooking³. These indicators reflect a household's access to modern, clean energy and water infrastructure. For instance, reliance on firewood or straw often signals the absence of natural gas pipelines, while coal usage, though inexpensive, contributes to pollution and suggests limited access to cleaner energy options. Similarly, using unsanitized water sources suggests insufficient piped or purified water infrastructure.

The affordability dimension is measured by combining household income and energy expenditure⁴ following the low-income, high-cost (LIHC) framework proposed by Hills (2012). This framework prevents the misclassification of high-income households as energy-poor. Under this framework, households are classified as low income if their income falls below the median within their county. Among these low-income households, those spending more than 10 percent of their income on energy costs are considered energy-poor from an affordability perspective. By incorporating both accessibility and affordability indicators, we capture the multifaceted challenges of household energy poverty, emphasizing not only the physical inaccessibility of modern energy and clean water, but also the financial burden associated with securing daily energy needs.

Table 1 presents the detailed components and corresponding weights of MEPI developed in this study. The selected indicators are widely used in prior research and are available in all waves of the 2012–2020 CEPS dataset. Earlier studies constructing an MEPI across multiple years often relied on indicators not consistently surveyed in each wave, requiring assumptions and imputations to address missing data (Li & Ma, 2023; Wang & Lin,

³ The cooking water is identified by the question “What kind of water does your family normally use for cooking?” in the CFPS Questionnaire. If the primary cooking water comes from river/lake water, well, rainwater, and cellar water/spring water, we assign cooking water =1; otherwise, cooking water =0. For households that rely on unsanitized water sources for cooking, their drinking water is typically free. These households are unable to access clean and secure drinking water. This situation reflects energy poverty from the perspective of accessibility, specifically the inability to obtain clean drinking water resources.

⁴ The energy expenditure includes electricity, heating, fuel, and water aspects, all of which are daily necessities.

2022)).⁵ In contrast, our approach avoids measurement errors by restricting the analysis to the indicators surveyed in each wave, thereby ensuring greater consistency and accuracy in measuring multidimensional energy poverty. For reference, a survey of all indicators used in previous studies is provided in Table A1.

Each of the three accessibility sub-indicators (firewood, coal, and cooking water), is assigned a weight of 1/6, contributing to half (1/2) of the total accessibility dimension. The affordability dimension accounts for the remaining half (1/2) using the LIHC indicator. This balanced weighting ensures equal emphasis on both the physical availability of clean energy and water and the financial capacity to afford these resources, are given equal emphasis in capturing the full spectrum of household energy poverty.

Table 1. MEPI Variable Description

Dimension	Indicator	Description	Reference	Weights
Accessibility	Firewood	=1 if the main cooking fuel is firewood/strew; otherwise =0	Du et al. (2022); Li and Ma (2023); Lin and Zhao (2021); Nie et al. (2021); Wang and Lin (2022); Yin et al. (2023); Zhang et al. (2019); Zhang et al. (2021)	1/6
	Coal	=1 if the main cooking fuel is coal; otherwise =0	Du et al. (2022); Li and Ma (2023); Lin and Zhao (2021); Nie et al. (2021); Wang and Lin (2022); Yin et al. (2023); Zhang et al. (2019); Zhang et al. (2021)	1/6
	Cooking water	=1 if the cooking water is directly from river/lake water, well, rainwater, cellar water/spring water; otherwise =0	Wang and Lin (2022)	1/6
Affordability	LIHC	=1 if energy expenditure > 10% of household income & household income < county median household income; otherwise =0	Cheng et al. (2021); Qin et al. (2022); Wang et al. (2022); Wang and Du (2024); Zhang et al. (2019)	1/2

⁵ For example, monthly average electricity usage and ownership of household appliances, such as TVs, computers, and mobile phones, were not included in the analysis to avoid introducing measurement errors, as these variables were not surveyed in every survey year and would have required imputation from previous surveys. Additionally, the year of construction could serve as a valuable measure for the MEPI, considering China's implementation of the Civil Building Energy Conservation Policy in 2008, which aims to enhance the energy efficiency of buildings. However, the CFPS dataset does not provide precise information on the year of construction of each household's residence. Although prior studies have commonly used the year of acquiring property rights for a house as a proxy for the construction year, the actual construction year data in this study were not sufficiently precise for inclusion.

Note: Energy expenditure includes electricity, heating, fuel, and water fee.

Table 2 presents the summary statistics for the outcomes and predictors used in this study. The predictor set includes variables commonly used in previous research on energy poverty. Prior studies have linked energy poverty primarily to household demographic and economic characteristics (Belaïd, 2018; Zhang et al., 2019). Specifically, we incorporate variables such as income (Abbas et al., 2022; Crentsil et al., 2019; Huang et al., 2022; Karpinska & Śmiech, 2021; Xie et al., 2022), education level (Alem & Demeke, 2020; Awan et al., 2022; Drescher & Janzen, 2021; Xie et al., 2022), family size (Mendoza et al., 2019), and residential area (Sharma et al., 2019; Xie et al., 2022). To ensure the robustness of median-based calculations, counties with fewer than 25 households were excluded from the analysis. Additionally, to mitigate the influence of extreme values, households in the top and bottom 2% for per capita income and per capita net assets were removed from the final sample.

Table 2 summarizes the key variables. The multidimensional energy poverty index (MEPI) was computed for 63,516 observations, yielding a mean of 0.21. On average, 71% of household members held an agricultural hukou. Economic status was measured using two county-relative ratios: the ratio of per capita income (mean = 1.39) and the ratio of per capita net assets (mean = 1.86), both exhibiting substantial variability, reflecting significant economic disparities. The mean age of the respondents was approximately 47 years, and adults in each household had an average of 7.56 years of education. The average family size was 3.72 persons, and the proportion of employed household members was 0.71. Approximately 92% of the sampled households were of *Han* nationality, and 92% of the household members had medical insurance. On average, the households have 42.53 square meters of living area per capita. Finally, 82% of households owned their homes.

Table 2. Descriptive Statistics

Variable	N	Mean	Std. Dev.	Min	Max
MEPI	63,516	0.21	0.25	0.00	0.83

Proportion of female household members	63,311	0.50	0.19	0.00	1.00
Proportion of household members with an agricultural Hukou	63,231	0.71	0.42	0.00	1.00
Ratio of per capita income to the county median	60,014	1.39	1.99	0.00	170.63
Ratio of per capita net assets to the county median	61,258	1.86	10.57	-1653.24	1318.29
Average age	63,302	46.79	12.68	18.00	95.00
Average years of education among adults in the household	63,142	7.56	3.91	0.00	22.00
Family size	63,516	3.72	1.87	1.00	21.00
Proportion of household members employed	61,595	0.71	0.38	0.00	1.00
<i>Han</i> nationality	62,709	0.92	0.25	0.00	1.00
Proportion of household members with medical insurance	63,217	0.92	0.21	0.00	1.00
Urban	62,446	0.50	0.50	0.00	1.00
Household living area per capita	60,308	42.53	58.01	0.25	4266.50
Home owned family	63,123	0.82	0.38	0.00	1.00

Source: Authors calculation from CFPS.

3. Methodology

3.1. Regression Approach to Identify the Relationship between MEPI and Predictors

This study employs a conventional regression framework, using MEPI as the dependent variable to assess how socioeconomic and demographic factors influence household energy poverty. Specifically, we regress MEPI on the set of explanatory variables from Table 2 while controlling for county- and year-level unobserved heterogeneity.

Formally, we estimate the Equation (1):

$$MEPI_{i,j,t} = \alpha + \beta X_{i,j,t} + \theta_j + \tau_t + \varepsilon_{i,j,t} \quad (1)$$

where $MEPI_{i,j,t}$ denotes the multidimensional energy poverty index for household i in country j and year t , $X_{i,j,t}$ represents the vector of explanatory variables from Table 2, θ_j and τ_t denote

county and year fixed effects, respectively, and $\varepsilon_{i,j,t}$ is the error term. To address potential within-county correlation, standard errors are clustered at the county level. The final regression sample comprised 8,719 households across 262 counties.

3.2. Machine Learning Technique to Identify Nonlinear Relationship

Although the conventional regression model in Section 3.1 enables us to estimate a linear relationship between MEPI and its predictors, it may not fully capture the complexity of real-world data. The relationship between MEPI and its predictors are often nonlinear and intricate. Moreover, linear models require the omission of observations with missing values, resulting in the use of only approximately 14% of the available sample. To address these limitations, we apply machine learning techniques that can detect nonlinearities in the data and handle missing values more effectively, and improve the accuracy and interpretability of results.

We estimate the relationship between MEPI and its predictors using a nonparametric gradient boosting technique originally developed by Friedman (2001). This technique sequentially updates prior base-learner models to minimize overall prediction error, improving accuracy and robustness against outliers. Unlike conventional regression models, gradient boosting enables us to uncover complex functional relationships that would otherwise be difficult to capture.

For implementation, we use eXtreme Gradient Boosting (XGBoost), a variant of gradient boosting (Chen et al., 2015), to estimate Equation (2), where $MEPI_{i,j,t}$ is the MEPI of household i in country j and year t and $X_{i,j,t}$ is the set of predictors of household i in country j and year t .

$$MEPI_{i,j,t} = f(X_{i,j,t}) \quad (2)$$

This method uses decision trees as base-learner models, and updates them sequentially. Notably, this method can handle missing values in predictors by treating them as distinct traits for estimation, rather than omitting them or requiring imputation. To optimize the

hyperparameters of XGBoost, each survey was randomly split into training and test sets in an 80:20 ratio. We then employed a 10-fold cross-validation and a grid search for hyperparameter tuning. Household-level sampling weights are used to fit the model.

In Section 3.1, the linear regression models provide interpretable coefficients, allowing us to directly assess predictor significance. However, nonlinear machine learning models, often function as a “black box,” making it difficult to understand how individual predictors influence MEPI (Leist et al., 2022; Murdoch et al., 2019).

To improve interpretability, a machine learning approach based on the SHAP technique proposed by Lundberg and Lee (2017). SHAP decomposes the model’s predictions, offering a clear view of how each predictor contributes to the predicted MEPI value, while controlling for all other factors. The SHAP value represents the individual contribution of a predictor in an observation relative to the model’s average prediction, with a positive value indicating a positive contribution, and a negative value indicating the opposite. A value of zero means that the predictor neither increases nor decreases the prediction from the average.

While SHAP values provide insight into individual predictor effects, they do not indicate which variables are most influential across the entire sample. To address this, we calculate feature importance scores--the mean absolute SHAP values for each predictor--This score parallels the significance checks in linear models, in which important variables are identified by examining the coefficient estimates and their standard errors. By contrast, in the nonlinear machine learning model, a higher feature importance score indicates that a variable has a greater overall impact on predicting the MEPI.

Although the feature importance score highlights how strongly a variable influences the MEPI, it does not explain the nature of the relationship, whether it is linear, nonlinear, or otherwise. For this purpose, we relied on partial dependence plots (PDPs) constructed from the SHAP values. These plots illustrate the marginal effect of a predictor on the MEPI while

controlling for all other factors, thus revealing how the predicted MEPI changes with variations in a particular predictor. By smoothing SHAP values for using a lowess function, these plots reveal complex or nonlinear patterns that linear models may not capture. This additional layer of interpretability complements the regression model from Section 3.1, offering a more nuanced understanding of how socioeconomic and demographic factors drive energy poverty in China.

4. Results

4.1. Linear Regression Results

The Ordinary Least Squares (OLS) regression results in Table 3 provide insights into how various predictors influence a household's MEPI. The findings indicate that a higher proportion of household members holding an agricultural hukou is positively and significantly associated with MEPI, suggesting that households with more agricultural hukou holders are at a higher risk of energy poverty. This result is consistent with Chang et al. (2024) who found that agricultural hukou households face greater barriers in transitioning to modern energy. These challenges stem primarily from limited access to modern energy infrastructure and financial constraints, which hinder the adoption of cleaner energy alternatives.

Similarly, an increase in the ratio of per capita household income to the county median is associated with a significant decline in MEPI, indicating that higher-income households are less likely to experience energy poverty. Wealthier households generally have greater financial capacity to afford modern energy and better access to energy infrastructure. This result aligns with previous studies (Cong et al., 2022; Igawa & Managi, 2022; Nguyen & Nasir, 2021), which also highlight the strong link between low household income and high energy poverty levels.

An increase in the average years of education among adults in the household is associated with a significant reduction in MEPI, reinforcing the idea that education facilitates

access to information and modern energy sources. This finding is consistent with prior research showing a negative relationship between education and energy poverty (Apergis et al., 2022; Banerjee et al., 2021; Makate, 2024). Education enhances human capital, equipping individuals with the knowledge and resources to transition from traditional fuels to cleaner energy options (Grubler, 2012).

Household size also exhibits a significant negative relationship with MEPI, suggesting that smaller households are more vulnerable to energy poverty. This result aligns with previous research (Abbas et al., 2020; Pachauri et al., 2004; Xie et al., 2022), which shows that larger households benefit from economies of scale in energy consumption. For example, Pachauri et al. (2004) found that a single-person household requires nearly three times more energy to prepare a meal than a six-member household, based on micro-survey data from India. This highlights the importance of household size in shaping energy consumption patterns, with smaller households facing lower energy efficiency and higher vulnerability to energy poverty.

Households located in urban areas tend to have significantly lower MEPI values than those in rural regions. This finding is consistent with earlier research, which emphasizes that urban settings benefit from better energy infrastructure and easier access to modern energy sources (Li et al., 2025).

Table 3. Regression for MEPI

	(1)
Proportion of female household members	0.017* (0.010)
Proportion of household members with an agricultural Hukou	0.085*** (0.009)
Ratio of per capita income to the county median	-0.036*** (0.009)
Ratio of per capita net assets to the county median	-0.000 (0.000)
Average age	0.002*** (0.000)
Average years of education among adults in the household	-0.007*** (0.001)
Family size	-0.016***

	(0.001)
Proportion of household members employed	-0.010*
	(0.005)
<i>Han</i> nationality	0.007
	(0.014)
Proportion of household members with medical insurance	-0.013
	(0.008)
Urban	-0.032***
	(0.007)
Household living area per capita	0.0001**
	(0.00005)
Home owned family	-0.002
	(0.005)
Constant	0.250***
	(0.029)
Observations	54,597
R-squared	0.282
Year FE	Yes
County FE	Yes

Notes: The dependent variable is MEPI. Standard errors are clustered at the county level and reported in parentheses. The symbols ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. All regressions include year and county fixed effects.

4.2. Feature Importance Scores

Table 4 presents the feature importance scores derived from the SHAP-based methodology, indicating that the ratio of per capita income to the county median is the most influential predictor of MEPI. This factor is followed closely by family size, average years of education among adults in the household, and urban residency. Notably, this ranking aligns with the OLS results in Table 3, which identified the same predictors as key determinants of household energy poverty. Additionally, the proportion of household members holding an agricultural hukou emerged as a moderately important factor, reinforcing its significant role in the linear model.

These findings are consistent with previous research on energy poverty determinants. Variables such as income (Abbas et al., 2022; Crensil et al., 2019; Huang et al., 2022; Karpinska & Śmiech, 2021; Xie et al., 2022), education level (Alem & Demeke, 2020; Awan et al., 2022; Drescher & Janzen, 2021; Xie et al., 2022), and family size (Mendoza et al., 2019)

have consistently been recognized as significant explanatory factors. By integrating traditional linear models and machine learning approaches, our study confirm the robust influence of socioeconomic and demographic characteristics on household energy poverty.

Table 4. Feature importance scores

	$E(SHAP)$
Ratio of per capita income to the county median	0.125
Family size	0.029
Average years of education among adults in the household	0.025
Urban	0.025
Proportion of household members with an agricultural Hukou	0.012
Ratio of per capita net assets to the county median	0.011
Average age	0.009
Household living area per capita	0.007
Home owned family	0.006
Proportion of household members employed	0.005
Proportion of female household members	0.004
Proportion of household members with medical insurance	0.003
<i>Han</i> nationality	0.002

Source: Authors calculation from CFPS.

4.3. Marginal Effects

This section analyzes the marginal effects of factors influencing multidimensional energy poverty in China using PDPs. While feature importance scores highlight the relative significance of predictors, they do not indicate whether these variables have positive or negative associations with energy poverty. To address this gap, we employ PDPs to visualize and interpret each predictor's contribution to energy poverty outcomes. The nonparametric machine learning approach enables us to detect nonlinear relationships, a feature that is often overlooked in linear models. Consequently, this approach provides a more nuanced understanding of the complex interactions between predictors and household energy poverty.

We focus on four policy-relevant variables that can be targeted to alleviate energy poverty: (1) the ratio of per capita income to the county median, (2) family size, (3) average years of education among adults in the household, and (4) the proportion of household members

with an agricultural hukou. Figure 1 illustrates the PDPs for these predictors based on the SHAP values smoothed using the Loess technique. A zero value on the y-axis indicates that the predictor does not influence the model's prediction, whereas positive or negative SHAP values imply that the predictor increases or lowers the predicted MEPI, respectively. The slope of the smoothing line is crucial in understanding how incremental changes in each predictor affect energy poverty outcomes.

[Insert Figure_1 near here]

Figure 1.A shows a steep negative slope in the SHAP value as the income ratio increases from zero to approximately one, after which the curve flattens. Households with income ratios close to zero exhibit highly positive SHAP values, indicating greater energy poverty risk, whereas households with income ratios above one experience flat or declining (near-zero or negative) SHAP values. Although Table 3 similarly identifies a negative coefficient for income, this PDP analysis reveals that income has its strongest impact on MEPI when the ratio is below the county median. However, beyond a ratio of one, additional income increases have diminishing effects on energy poverty reduction.

Figure 1.B reveals a U-shaped relationship between family size and MEPI. From zero to about 10 members, the SHAP value decreases steeply, reaching its lowest point at family size 10. For smaller households, (<3 members), the SHAP value is strongly positive, implying a higher energy poverty risk. As family size grows towards 10, the SHAP value dips below zero, signaling a reduced risk. Beyond 10 members, the curve rises again, though at a less steep rate than. While the OLS coefficient for family size in Table 3 is negative, the PDP clarifies that the most significant reduction in energy poverty occurs for households growing from very small sizes to around 10 members, beyond which the effect begins to reverse.

Figure 1.C shows that households with very low education levels (close to zero) have substantially higher predicted MEPI values, as reflected by the strongly positive SHAP values

at the left edge of the plot. The SHAP value then declined steeply from approximately three to nine years of education, crossing below zero near the midpoint, suggesting that moderate education levels had already begun to mitigate energy poverty. Beyond nine years, the slope remains negative but becomes flatter, indicating that further increases in education still reduce the energy poverty risk, but with diminishing marginal returns. While the OLS model in Table 3 confirms a negative coefficient for education, the PDP reveals nonlinear effects that linear models fail to capture.

Finally, Figure 1.D reveals an upward-sloping relationship between the proportion of household members with an agricultural hukou and the predicted MEPI from 0 to approximately 0.8, after which the slope flattens noticeably. When the hukou proportion is near zero, the SHAP value is negative, implying a lower energy poverty risk; however, as the proportion increases toward 0.8, the SHAP value steadily rises, indicating a higher energy poverty risk. Beyond 0.8, further increases in the proportion of agricultural hukou holders have little additional effect on MEPI. Although Table 3 shows a significantly positive coefficient for agricultural hukou status, the PDP provides further insights, demonstrating that the greatest adverse impact occurs between 0 and 0.8, after which the effect stabilizes.

5. Conclusion

This study examined the determinants of household energy poverty in China from 2012 to 2020 using a consistent MEPI framework and a nonlinear machine learning approach. By relying on indicators that were consistently surveyed across CFPS waves, we avoided measurement errors associated with imputed or missing data. The resulting MEPI incorporates both accessibility and affordability, providing a robust depiction of China's evolving energy poverty landscape.

Our findings contribute to the broader literature on energy poverty by combining traditional regression models with machine learning techniques. The OLS model confirmed that socioeconomic factors play a central role in shaping household energy poverty.

Specifically, higher household income, larger family size, higher education levels, and urban residency (lower proportion of agricultural hukou holders) are associated with significantly lower energy poverty. Thus, energy poverty remains concentrated among poorer, smaller, less-educated, and rural households, a pattern consistent with broader vulnerability findings in China. The machine learning approach reinforces these findings while offering additional insights. Using XGBoost models and SHAP values, we identify household income, family size, education, and hukou status as the most influential predictors of energy poverty. Importantly, the ML model reveals complex nonlinear relationships that the linear regression model fails to fully capture, providing a deeper understanding of how these factors contribute to energy deprivation.

These findings highlight the importance of considering both affordability and infrastructural access when measuring and addressing energy poverty in China. The nonlinear insights gained from XGBoost are particularly relevant for policymakers. A purely linear framework may overlook critical tipping points, such as income thresholds or shifts in the impact of family size. In contrast, the machine learning approach uncovers how even modest increases in education or household size can significantly reduce energy poverty within specific ranges. Consequently, targeted policies that prioritize education, expand rural clean energy infrastructure, and support smaller households could be more effective than one-size-fits-all interventions. By applying a consistently measured MEPI and leveraging nonlinear methodologies, this study enhances our understanding of household energy poverty, offering empirical insights and actionable policy guidance for future interventions in China.

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Data Availability: The China Family Panel Survey (CFPS) data used in this study can be accessed through the Peking University Open Research Data Platform after completing the registration process. Detailed instructions for registration are available at the following link: <https://cfpsdata.pku.edu.cn/#/home>

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Appendix Table

Table A1. Indicators, Studies, and CFPS Waves Used in Previous Research

Indicators	Study (Year)	CFPS Waves Used
(1) Fuel		
Firewood	Du et al. (2022)	2014, 2016, 2018
	Hong et al. (2022)	2012, 2014, 2016, 2018
	Lei and Xu (2025)	2012, 2014, 2016, 2018, 2020, 2022
	Li and Ma (2023)	2016, 2018
	Li et al. (2025)	2014, 2016, 2018
	Li et al. (2023)	2014, 2016, 2018
	Li et al. (2022)	2014, 2016, 2018
	Lin and Zhao (2021)	2018
	Nie et al. (2021)	2010, 2012, 2014, 2016, 2018
	Qin et al. (2022)	2014, 2016, 2018
	Wang et al. (2022)	2012, 2014, 2016
	Wang and Du (2024)	2012, 2014, 2016, 2018
	Wang and Lin (2022)	2014, 2016, 2018
	Yan et al. (2024)	2012, 2014, 2016, 2018
	Yin et al. (2023)	2014, 2016, 2018
	Zhang et al. (2019)	2012, 2014, 2016
	Zhang et al. (2021)	2014, 2016, 2018
Coal	Du et al. (2022)	2014, 2016, 2018
	Li et al. (2025)	2014, 2016, 2018
	Li and Ma (2023)	2016, 2018
	Lei and Xu (2025)	2012, 2014, 2016, 2018, 2020, 2022
	Li et al. (2023)	2014, 2016, 2018
	Li et al. (2022)	2014, 2016, 2018
	Lin and Zhao (2021)	2018
	Nie et al. (2021)	2010, 2012, 2014, 2016, 2018
	Qin et al. (2022)	2014, 2016, 2018
	Wang et al. (2022)	2012, 2014, 2016
	Wang and Du (2024)	2012, 2014, 2016, 2018
	Wang and Lin (2022)	2014, 2016, 2018
	Yan et al. (2024)	2012, 2014, 2016, 2018
	Yin et al. (2023)	2014, 2016, 2018
	Zhang et al. (2019)	2012, 2014, 2016

	Zhang et al. (2021)	2014, 2016, 2018
Bio gas	Yin et al. (2023)	2014, 2016, 2018
(2) Energy expenditure		
Electricity fee	Cheng et al. (2021)	2012, 2014, 2016, 2018
	Du et al. (2022)	2014, 2016, 2018
	Gu (2023)	2016, 2018
	Lei and Xu (2025)	2012, 2014, 2016, 2018, 2020, 2022
	Li et al. (2025)	2014, 2016, 2018
	Li et al. (2023)	2014, 2016, 2018
	Li et al. (2022)	2014, 2016, 2018
	Lin and Zhao (2021)	2018
	Nie et al. (2021)	2010, 2012, 2014, 2016, 2018
	Qin et al. (2022)	2014, 2016, 2018
	Shi et al. (2023)	2014, 2016, 2018
	Wang et al. (2023)	2012, 2014, 2016, 2018
	Wang et al. (2022)	2012, 2014, 2016
	Wang and Du (2024)	2012, 2014, 2016, 2018
	Wang and Lin (2022)	2014, 2016, 2018
	Yan et al. (2024)	2012, 2014, 2016, 2018
	Yin et al. (2023)	2014, 2016, 2018
	Zhang et al. (2019)	2012, 2014, 2016
Fuel fee, Heating fee	Cheng et al. (2021)	2012, 2014, 2016, 2018
	Du et al. (2022)	2014, 2016, 2018
	Gu (2023)	2016, 2018
	Lei and Xu (2025)	2012, 2014, 2016, 2018, 2020, 2022
	Li et al. (2025)	2014, 2016, 2018
	Li et al. (2023)	2014, 2016, 2018
	Li et al. (2022)	2014, 2016, 2018
	Lin and Zhao (2021)	2018
	Nie et al. (2021)	2010, 2012, 2014, 2016, 2018
	Qin et al. (2022)	2014, 2016, 2018
	Shi et al. (2023)	2014, 2016, 2018
	Wang et al. (2023)	2012, 2014, 2016, 2018
	Wang et al. (2022)	2012, 2014, 2016
	Wang and Du (2024)	2012, 2014, 2016, 2018
	Wang and Lin (2022)	2014, 2016, 2018
	Yan et al. (2024)	2012, 2014, 2016, 2018
	Zhang et al. (2019)	2012, 2014, 2016

Water fee	Cheng et al. (2021)	2012, 2014, 2016, 2018
	Qin et al. (2022)	2014, 2016, 2018
	Wang et al. (2022)	2012, 2014, 2016
	Wang and Du (2024)	2012, 2014, 2016, 2018
	Zhang et al. (2019)	2012, 2014, 2016
(3) Income		
Income (yearly)	Du et al. (2022)	2014, 2016, 2018
	Gu (2023)	2016, 2018
	Lei and Xu (2025)	2012, 2014, 2016, 2018, 2020, 2022
	Lin and Zhao (2021)	2018
	Nie et al. (2021)	2010, 2012, 2014, 2016, 2018
	Qin et al. (2022)	2014, 2016, 2018
	Shi et al. (2023)	2014, 2016, 2018
	Wang et al. (2023)	2012, 2014, 2016, 2018
	Wang et al. (2022)	2012, 2014, 2016
	Wang and Lin (2022)	2014, 2016, 2018
	Zhang et al. (2019)	2012, 2014, 2016
(4) Cooking water		
	Wang and Lin (2022)	2014, 2016, 2018
(5) Electricity supply		
	Lei and Xu (2025)	2012, 2014, 2016, 2018, 2020, 2022
	Wang and Lin (2022)	2014, 2016, 2018
	Zhang et al. (2021)	2014, 2016, 2018
(6) Home appliance expenditure		
	Wang and Lin (2022)	2014, 2016, 2018
(7) House maintaining expenditure		
	Wang and Lin (2022)	2014, 2016, 2018
(8) Home appliance		
Mobile phone, TV	Wang et al. (2023)	2012, 2014, 2016, 2018
	Yin et al. (2023)	2014, 2016, 2018
	Zhang et al. (2021)	2014, 2016, 2018
Refrigerator	Wang et al. (2023)	2012, 2014, 2016, 2018
	Zhang et al. (2021)	2014, 2016, 2018
Internet- connected computer usage	Li and Ma (2023)	2016, 2018
	Yin et al. (2023)	2014, 2016, 2018

Air cleaning facility ownership, Mobile phone usage	Li and Ma (2023)	2016, 2018
Air conditioners, Bicycle, Car, Motorbike, Washing machines, Ratio	Wang et al. (2023)	2012, 2014, 2016, 2018
(9) Others		
Household appliance value	Yin et al. (2023)	2014, 2016, 2018
Monthly electricity usage (kWh)	Wang et al. (2023)	2012, 2014, 2016, 2018
Year of construction	Wang et al. (2023)	2012, 2014, 2016, 2018